

GISS 2026

Sisyphus Optimizing in a **Dynamic Universe**

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FIGURE

Sisyphus — boulder mid-roll, *downhill*. He's below it,
holding it back.

Camus got one thing wrong.
The **direction**.

The boulder rolls down. He goes with it.

AI's newest idea is
your brain's **oldest**.

① DISTANCE — ONE IDEA, THREE COSTUMES

 ASTRONOMY

redshift picks your rung.

 AI

the embedding *is* "close."

 PERSON

too far: a stranger.
too close: only flaws.

sampling → resolution → distance

Nyquist binds all three

past the band → hallucinate · below it → forget

⌚ LOSS — IT FALLS OUT OF THE DISTANCE

once "close" is defined, the loss writes itself.

But choosing the distance
was **itself** an optimization.

loops inside loops

© DESCEND · POLYAK 1964 — THE "HEAVY BALL METHOD"

Mass is **memory**.

Downhill, **gravity switches sides**.

pushing → restraining · let go = divergence

④ MODEL = FORGET

To model is to **forget**.

hallucination $\leftrightarrow$ amnesia

which pixels are physics — and which are the model's?

J A I S P · R U B I N - E U C L I D F O U N D A T I O N M O D E L · H E M M A T I E T A L .

Forgetting **with a hedge.**

freeze the encoder — build a good model, then leave it alone.

0.45 mag **50→14 mas** **0×**
deeper astrometry retraining, new field

HERO FIGURE

one source, ten bands → one frozen representation.

FIGURE — BIG & GORGEOUS
a stunning super-resolved galaxy.

Sharper — but real?

shape: worse than a blur · spectra: 26% real · photometry: biased

Optimizing **every second?**
Something is wrong.

a healthy mind coasts. re-descend only on: distrust · surprise · a new question.

exploding → anxiety · **vanishing** → depression

A paper is a photograph of
the boulder **mid-slope**.

midway is the only place one can be written — *"done, for now."*

a real win — then you move: a new direction opens, or the well goes **flat**.

© THE TARGET MOVES

You reach — and it **recedes**.

that isn't Sisyphus. that's *Tantalus* — water retreating from his lips, fruit the wind lifts from his hand.

the target moves **because** you reached.

Goodhart's law · reward hacking · the alignment problem

© THE TARGET MOVES

You reach the bottom —
the valley **splits**.

$H_0 \rightarrow$ the Hubble tension · intimacy has a Nyquist limit

you see me at podium resolution — which I've selected carefully.

GALAXY real clump, or the PSF?

NEURAL NET physics, or prior?

BRAIN memory, or a reconstruction?

US you, or my projection?

measurement, or invention.

The receding target
was never the **punishment**.

reach the fruit and the wanting ends · no gradient → nothing moves → heat death.

"it's coming home" · 60 years

فراق · Hafez · 700 years

Tantalus, the ghazal, the stadium — *the distance is the wanting*.

The Glorious Gradient

joy, sadness — or the joy inside the sadness.

The sweetness was never at the bottom.
It was on the **slope**.